

Internet & Web Fundamentals

Revised Study Notes — Concepts, Protocols & Client-Server Communication

Section	Covers
1	The Internet & the World Wide Web
2	How Data Actually Travels — TCP/IP & DNS (new)
3	Browsers and Search Engines
4	Internet Protocols — HTTP, HTTPS, FTP + SMTP/SSH (new)
5	Client-Server Model and URL Anatomy (new)
6	HTTP Requests & Responses
7	GET vs POST
8	HTTP Status Codes (expanded list)
9	Where HTML/CSS/Bootstrap/Tailwind fit in
10	Revision Quiz

1. The Internet & the World Wide Web

1.1 What is the Internet?

Definition: The Internet is a massive, global network made up of millions of smaller networks, all connected together and able to exchange data using a shared set of rules (protocols). No single company or country owns it — it's cooperative infrastructure.

Example: Think of the Internet as a city's underground pipe-and-cable network — the physical wiring that could carry water, electricity, or gas. On its own, it's just infrastructure. What flows through it is a separate matter.

1.2 What is the World Wide Web?

Definition: The Web is one particular service that runs on top of the Internet — a huge collection of linked documents (webpages) that you view through a browser using HTTP/HTTPS.

Example: If the Internet is the pipe network, the Web is the water flowing through certain pipes. Email and video calls are different things flowing through the same pipes.

Internet	World Wide Web
Global network of interconnected devices	A service of linked documents/pages
Infrastructure layer	Application layer built on that infrastructure
Also carries email, gaming, video calls, FTP	Specifically uses HTTP/HTTPS

2. How Data Actually Travels — TCP/IP & DNS

This section wasn't in the original manual but helps explain how a request even reaches the right computer.

2.1 TCP/IP in brief

Definition: TCP/IP is the core rulebook the Internet runs on. IP (Internet Protocol) gives every device an address and handles routing packets of data to it. TCP (Transmission Control Protocol) makes sure those packets arrive complete, in order, and error-free — re-sending anything lost along the way.

Example: Streaming a video on Hotstar breaks the video into thousands of small packets; IP routes each one, TCP reassembles them in the correct order on your device.

2.2 DNS — the Internet's phonebook

Definition: The Domain Name System (DNS) converts a human-readable address like `www.amazon.in` into the numeric IP address (e.g. `205.251.242.103`) that computers actually use to locate the server.

Example: Typing `www.wikipedia.org` into a browser triggers a DNS lookup behind the scenes before the actual page request is even sent.

Recap / Extra: Browser needs an address -> DNS resolves the name to an IP -> TCP/IP delivers the actual request/response packets -> HTTP defines what those packets mean.

3. Browsers and Search Engines

3.1 Web Browser

Definition: An application that requests pages from servers and renders their HTML, CSS, and JavaScript into something visible and interactive.

Browser	Rendering Engine	JS Engine
Google Chrome	Blink	V8
Microsoft Edge	Blink	V8 (Chakra retired)
Mozilla Firefox	Gecko	SpiderMonkey
Apple Safari	WebKit	JavaScriptCore

Recap / Extra: This rendering-engine split is why the exact same website can occasionally look or behave slightly differently across browsers.

3.2 Search Engine

Definition: A website that crawls and indexes other websites so users can find information by typing keywords.

Aspect	Browser	Search Engine
What it is	An installed application	A website
Example	Firefox	DuckDuckGo
Job	Lets you visit any URL	Helps you discover a URL to visit

4. Internet Protocols

Definition of a protocol: An agreed set of rules two systems follow so their communication is understood on both ends — similar to how two people need a shared language and grammar to have a conversation.

Protocol	Full Form	Port	Purpose
HTTP	HyperText Transfer Protocol	80	Transfers webpages; not encrypted
HTTPS	HTTP Secure	443	Encrypted HTTP using SSL/TLS
FTP	File Transfer Protocol	20/21	Uploads/downloads files to/from a server
SMTP*	Simple Mail Transfer Protocol	25	Sends email between mail servers
SSH*	Secure Shell	22	Encrypted remote login to a server

Recap / Extra: * SMTP and SSH are additional protocols beyond the original notes — included since they're commonly asked about alongside HTTP/FTP.

4.1 HTTP vs HTTPS

HTTP	HTTPS
No encryption — data readable in transit	Encrypted via SSL/TLS certificate
Port 80	Port 443
Example: <code>http://example.org</code>	Example: <code>https://sbi.co.in</code> (banking needs this)

Example: A banking or payment site without HTTPS would let anyone on the same network read your card details as they travel — this is why browsers actively warn users about non-HTTPS login pages.

4.2 FTP

Definition: Used to move files between a local computer and a remote server — most often for publishing website files to a hosting server.

Example: Common tools: FileZilla, WinSCP, or the built-in FTP client in most code editors.

5. Client-Server Model & URL Anatomy

Term	Definition	Example
Client	The device/software requesting data	Your phone's browser
Server	The computer that stores and returns the requested data	Amazon's web servers

5.1 Anatomy of a URL (new)

Breaking down <https://www.wikipedia.org/wiki/Internet> piece by piece:

Part	Value	Meaning
Scheme	https://	Protocol used to fetch the resource
Host/Domain	www.wikipedia.org	Which server to contact
Path	/wiki/Internet	Which resource on that server

6. HTTP Requests and Responses

HTTP Request: A message the client sends asking the server for a resource.

```
GET /wiki/Internet HTTP/1.1
Host: www.wikipedia.org
User-Agent: Mozilla/5.0
```

Example: Plain English: "Please send me the Internet article page."

HTTP Response: The server's reply, including a status line, headers, and body.

```
HTTP/1.1 200 OK
Content-Type: text/html

<html>...</html>
```

Example: Plain English: "Here is the page you asked for, and it loaded fine."

7. GET vs POST

The original notes used a login form for POST — here's a different, equally common example: an online checkout form.

Aspect	GET	POST
Purpose	Fetch/read data	Submit/send data
Data location	In the URL as a query string	In the request body
Example	https://shop.in/search?item=shoes	Submitting a checkout form (address, payment info)
Visible in URL/history?	Yes	No
Cacheable/bookmarkable?	Yes	No
Security	Less secure for sensitive data	More secure — used for sensitive submissions

8. HTTP Status Codes — Expanded List

The original notes covered the most common codes. A few additional ones are marked (new) below, since they come up often in real APIs.

Code	Name	Meaning	Example
100 (new)	Continue	Server got the initial part of the request	Large file upload in progress
200	OK	Request succeeded	A news article page loads normally
201	Created	New resource created	New account signed up on an app
204 (new)	No Content	Success but nothing to return	A "like" button click with no page reload
301	Moved Permanently	Resource has a new permanent address	An old blog URL redirects to a new one
302	Found	Temporary redirect	Redirected to a login page first
400	Bad Request	Malformed request	A search box submitted with invalid characters
401	Unauthorized	Login/authentication required	Opening a Gmail inbox while logged out
403	Forbidden	Authenticated, but not permitted	A regular user trying to open an admin dashboard
404	Not Found	Resource doesn't exist	A mistyped product page URL on Flipkart
429 (new)	Too Many Requests	Rate limit exceeded	Refreshing a ticket-booking page too fast
500	Internal Server Error	Server-side crash/bug	A bug in the site's backend code
503	Service Unavailable	Server temporarily overloaded/down	IRCTC website during Tatkal booking rush

Recap / Extra: Quick way to remember the ranges: 1xx = please wait, 2xx = all good, 3xx = look elsewhere, 4xx = you (the client) made a mistake, 5xx = the server made a mistake.

9. Where HTML, CSS, Bootstrap and Tailwind Fit In

This connects the protocol/network side of things back to what you're actually building in the web design labs.

Layer	Technology	Role
Structure	HTML	Defines the content and structure of a page
Presentation	CSS	Controls colours, spacing, fonts, layout
Pre-built styling	Bootstrap	Ready-made responsive components (grids, navbars, buttons)
Utility-first styling	Tailwind CSS	Small utility classes composed directly in HTML

Example: How it connects to Section 6: when a browser sends a GET request for a webpage, the response body is the HTML — which then pulls in CSS/Bootstrap/Tailwind stylesheets as separate requests, and any JavaScript, before the browser finishes rendering the full page.

10. Revision Quiz

1. What is the difference between the Internet and the World Wide Web?
 2. What role does DNS play when you type a website address into a browser?
 3. Name two responsibilities of TCP and IP respectively.
 4. Why is HTTPS considered safer than HTTP for a banking website?
 5. What is the difference between a browser and a search engine, with one example each?
 6. Which port does FTP use, and what is it typically used for?
 7. Write a sample HTTP request line for fetching a page called /about.html.
 8. Give one real-world scenario where a GET request would be used, and one for POST.
 9. What does a 401 status code mean, and how is it different from a 403?
 10. Why might a booking website return a 503 status code during a high-traffic sale?
 11. In the HTML/CSS/Bootstrap/Tailwind stack, which one controls appearance directly through HTML classes?
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